Product Management

Applying PLM, DCAM, SLM, and SCM to our evolving Data Supply Chain

May, 2021

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Product Management (Data)

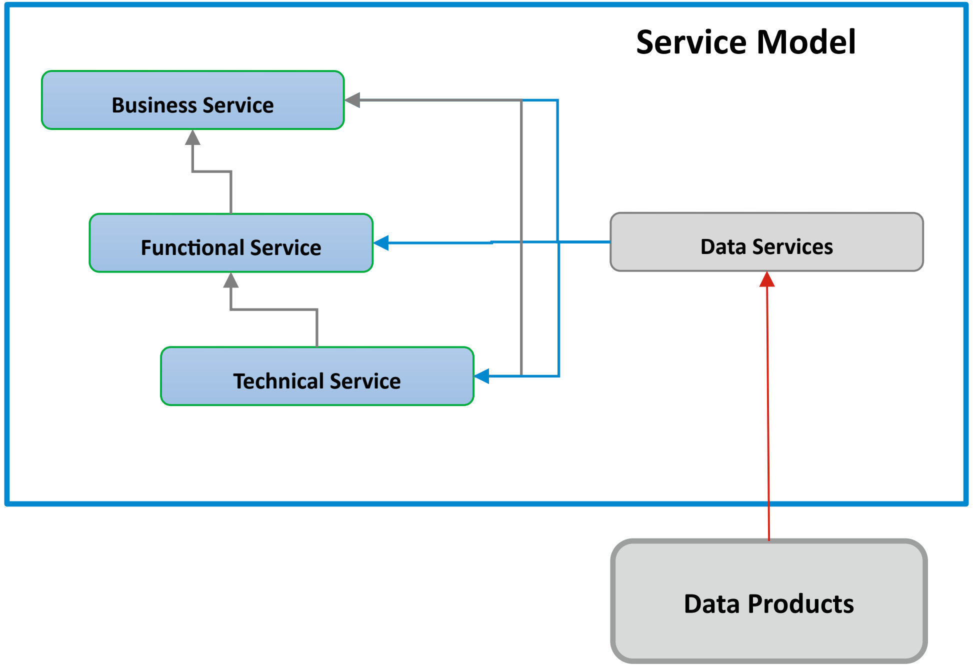
- Organizing Knowledge
- Developing the Right Products
- Delivering Perfect Orders

Developing the Right Products: Value-Driven Design

- Everything we create and deliver with our Data Factory needs to start with a clear understanding of value. Clarifying what consumers (stakeholders) want is the start for measuring value. Stakeholders have a set of desired outcomes that they are trying to achieve when they perform a job (diagnosing a line outage, planning a new tower, replacing an antenna array, or installing a new power feed for a tower).
- When we provide Data Products, we need to design the product to deliver the right outcomes to help our stakeholders perform their jobs (more easily, faster, with less chance of making mistakes, and with less stress and frustration).
- A precise understanding of the job steps they are performing is key to fully understanding their desired outcomes and guiding our requirements and design specifications with precision.

- All of this content is data that we need to capture within the product development lifecycle so we can provide a clear traceability from stakeholder thru design thru development, testing, certification, and operational support.

The Lineage of Value Discovery



- Stakeholders perform and/or rely upon one or more services to get their jobs done.
- Typically, access to information is provided thru one or more system that provide Technical Services. These systems provide content within a business and functional context.
- In some cases, with reports, dashboards, or analytics stakeholders may directly consume data services.
- Data services providers require access to the right data, which is accurate and properly curated (quality, enriched with relevant context and semantics) so they can deliver the right outcomes for stakeholders.
- This is where we trace the desired outcomes back to the Data Products we need to design, deliver, and support (operationally).

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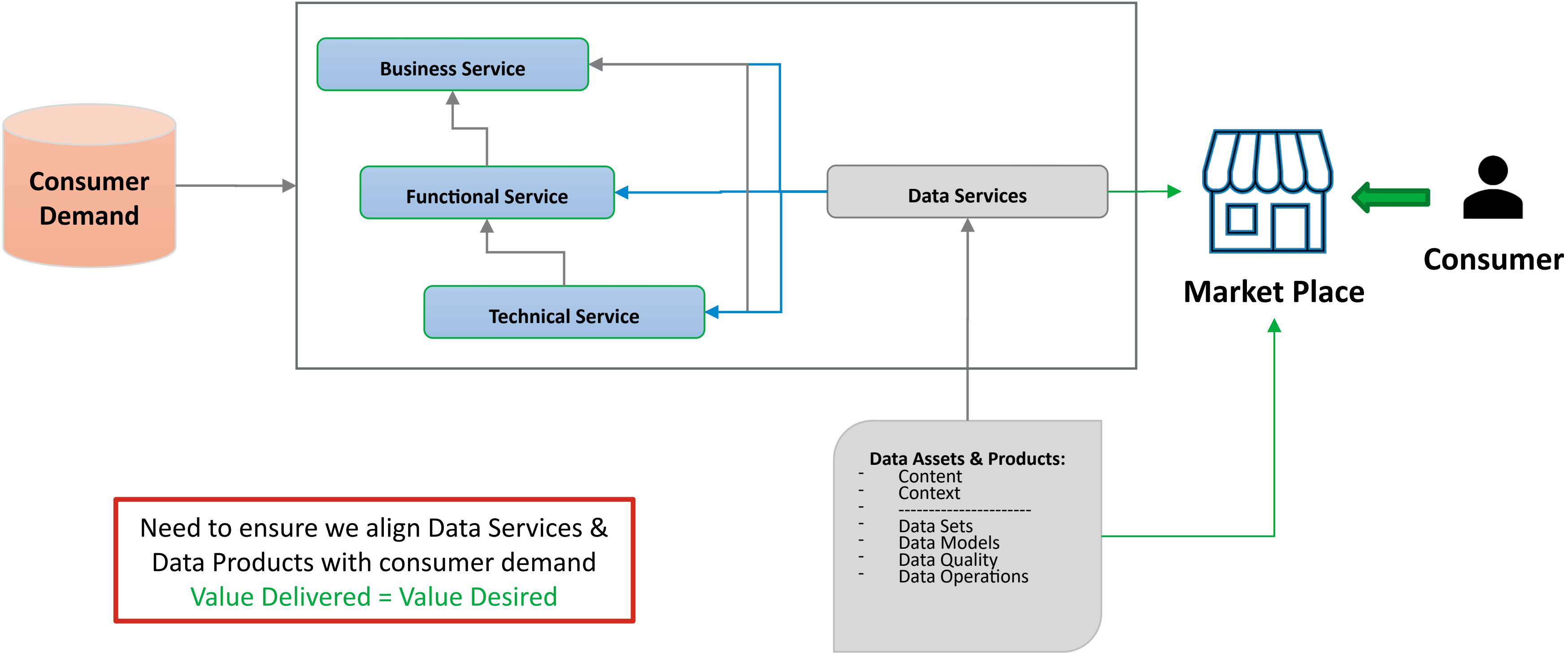
Value-Driven Lineage: Demand thru Data Products



Need to ensure we align Data Services & Data Products with consumer demand
Value Delivered = Value Desired

Product Lifecycle Management (PLM)

Product Design

Discover → Define → Design → Develop → Deliver

Version & Configuration Management

Product Transfer

Certify & Publish

Release Management

Product Deliver

Step1 Acquire → Step 2 Profile → Step 3 Transform → Step 4 Certify → Step 5 Publish

Product Support

Operations Management
(monitor, triage, investigate, report, support)

Issue Management

Governance needs to span across the full product lifecycle. Goal is to promote the 5 C's (Compliance, Collaboration, Cooperation, & Communication, & Control)

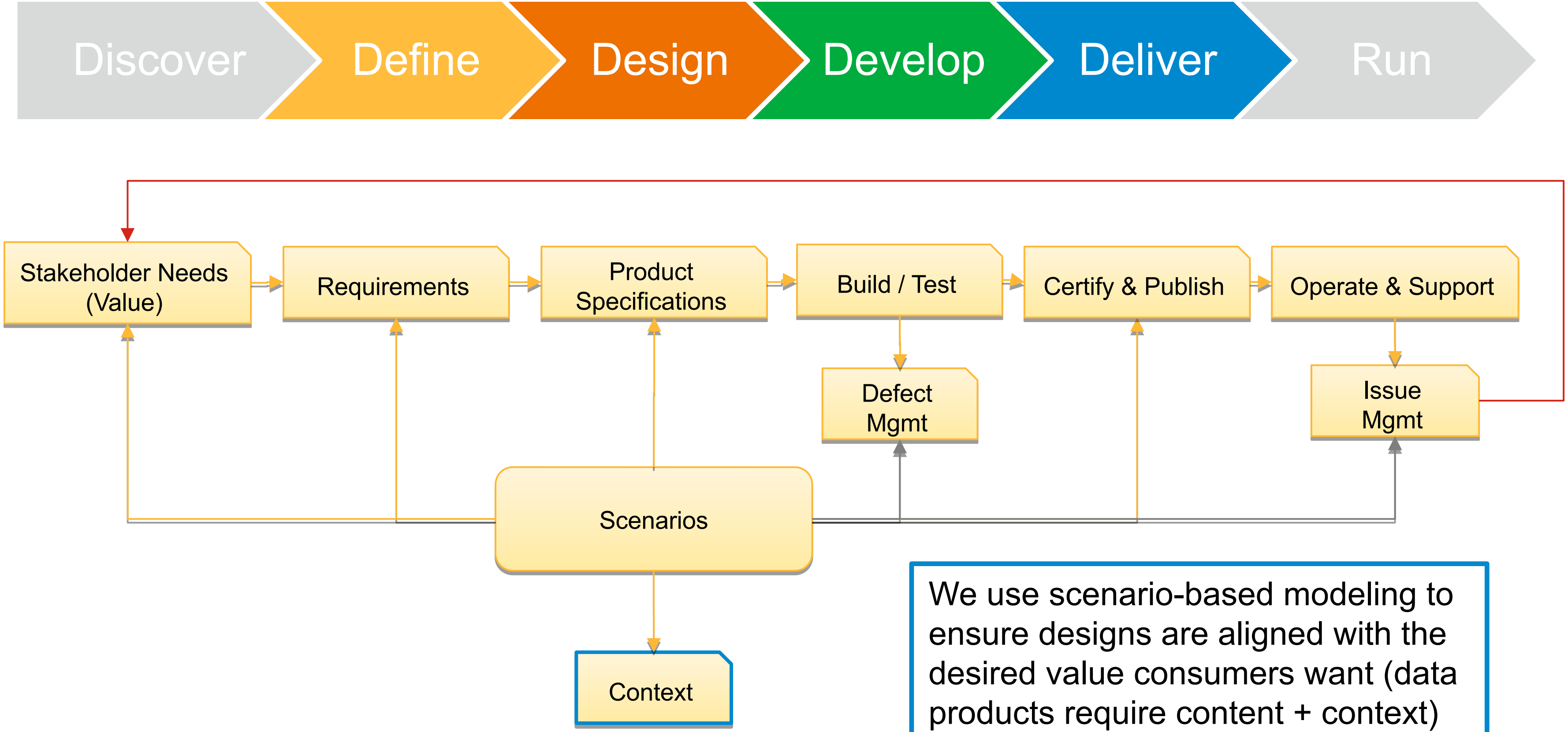
Everyone's goals should be oriented towards value delivery for all stakeholders:

- Consumers
- Operations
- Providers
- Partners
- Policy

Different Building Blocks for Data Products

- Data Products are uniquely different than physical products since they are comprised of virtual components. These components require formal structure and organization to ensure data products are designed and assembled consistently and accurately.
- Data Products are built atop knowledge artifacts so we need a system to organize these to create meaning and support flexibility and adaptability as digital products are not bounded by physical constraints and properties.
- We'll borrow from existing disciplines to help us design, build and operate our digital factor to create our digital products:
  - W3C – World Wide Web Consortium for organizing our knowledge
  - PDMA - Product Development Mgmt. Association for designing & building data products
  - ASCM – Association for Supply Chain Mgmt. for designing our data factory & running operations

Designing and Developing our Data Product: Product Lifecycle Mgmt.



We use scenario-based modeling to ensure designs are aligned with the desired value consumers want (data products require content + context)

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Delivering Perfect Orders Requires Perfect Planning

- Data Products consist of content (curated structured and/or unstructured data) plus context (addition of metadata and semantics to provide meaningful and understandable relevancy).

- Creating perfect orders for any product requires 2 things:

    - **Process Plan** – This is a recipe which has all of the steps required to acquire, make, and deliver the product perfectly. All ingredients that go into the making of the product and the specific components (equipment, software, data, services, models, etc.) including their version and configurations are all referenced in the Process Plan.

    - **Bill of Materials** – (BoM) This is the list of ingredients that must be prepared and qualified to meet the precise standards set forth in the Process Plan in order to create a perfect order.

Creating Perfect Orders: Process Plan & BoM

Discover → Define → **Design → Develop → Deliver** → Run

Content

Data Inputs

Context

Vocabulary(s)

Taxonomy(s)

Ontologies

- Business Rules
- Measures
- Models (Assets)
- Models (Data Structures)
- Logic (Transforms)
- Logic (Mapping & Merging)
- …

Data Services

- Profiling, Quality, Management
- Discovery, Consumption
- Security, Traceability, Lineage
- Orchestration, Transforms
- Registry, Catalog
- Promotion, Publishing

Bill of Materials (BoM)
(List of all inputs AND components in the Catalog)
Content + Context

Process Plan
(Script used to orchestrate the building of a data product)
Data Acquisition thru Data Product Publishing

Delivery Lifecycle: Data Product

Step1	Step 2	Step 3	Step 4	Step 5
Acquire	Profile	Transform	Certify	Publish

All products are built and delivered using an automated script (Process Plan).

Bill of Materials (List of Ingredients) + Process Plan (Recipe to Build a Data Product)

- Every data product has a unique Process Plan
- Each version of a data product has a unique Process Plan
- Quality is built into each step in the Process Plan
- The Process Plan provides E2E orchestration to build, certify, & deliver (publish) a data product
- A Certification Script is built as an adjunct to verify each instance of a data product in production

Blending Established Disciplines into a Digital Supply Chain

Discover → Define → Design → Develop → Deliver → Run

Product Management is responsible for maturing and blending the following industry standard disciplines to support:

- **PLM – Product Lifecycle Management**
  - Drive Design Thinking and Quality by Design for Marketplace (Digital Supply Chain) and all of the products and services we deliver for consumers.
- **SCM – Supply Chain Management**
  - Marketplace is a digital supply chain for consumers to Discover, Search, and Order data assets, data products, digital twins, and Service Requests (work orders) to help them get their jobs done Right
- **DLM – Data Lifecycle Management**
  - Applying the DCAM discipline across all digital assets and their lifecycles
- **SLM – Service Level Management**
  - Introduce the Provider, Producer, and Consumer model and work with teams involved in creating and delivering services to Creating a COE for service management across lifecycle delivery and operations (run)
- **SE – Systems Engineering**
  - Systems Engineering is complimentary to all the above disciplines and ties them together along with other software engineering and architecture disciplines

Operational Planning: Transitioning Product from Design to Delivery

- Transfer planning from Product Design (incremental) to Product Delivery (ongoing) requires careful planning to ensure the handoff is done right and operational teams are prepared to support stakeholder needs and manage expectations.

- Before any data product is put into production, it needs to be Certified. Initially, this will be a somewhat manual effort, but we will need to develop scripts to automate the certification process to support more rapid deployment and to keep pace with continuous improvement.

- Before a data product can be transitioned into production, we need to ensure we can support it and diagnose any issue by leveraging the complete set of data and knowledge artifacts used during product design and any upstream changes to data assets (inputs) and their source system changes (lineage).

Product Transfer: Publishing & Release Management

Product Transfer

Certify & Publish

Release Management

Paradigm Shift: Need to shift from Document to Data-Centric Planning & Design (everything is data in a digital product world, and everything should be traceable to data not documents)

All Design Artifacts

- Customer needs (Value)
- Requirements (Use Cases)
- Design Specifications for all inputs & components
- Deliver (Build & Test)
- Test Harnesses used as input for Cert Script
- Value Cert (trace from Value thru Product Design)
- Defects captured
- Changes captured as data artifacts

All Deliver Artifacts

- Bill of Materials
- Process Plan
- Operations Transfer
  - Fit for Purpose
  - Scope of Use & Support
  - Support Plan & Escalation
  - Investigation Process & tools
  - Protection & Provisioning
  - Certification Plan & Results
  - Quality Assessment
  - Security Assessment
  - Release & Impact Update
  - Operations Acceptance*

* Separate layer of skilled operations & technology staff

All artifacts are versioned and captured as data (not docs).

Policy is part of requirements and design. (includes Security)

Product design is not limited to what gets produced (data sets).

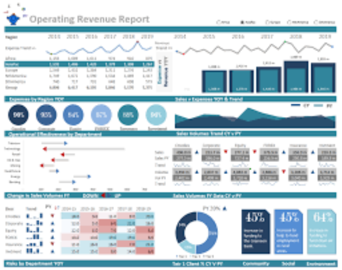
Design & Deliver needs to span the full lineage of data product through customer experience.

Product Operations: Production Operations & Consumer Support

Product Support

Operations Management
(monitor, triage, investigate, report, support)

Monitor & Measure



- Measurement
- SLE / SLA Review
- Pipeline Mgmt.
- Trending & Perf. Mgmt
- Consumer Satisfaction
- Value Delivery Measures

Manage

Change Management

- New products
- Existing products
- Communication
- Release Planning

Issue Management

- Triage & RCA
- Problem Mgmt.
- Resolution Mgmt.
- Quality Measures

Consumer Management

- Portfolio Management
- SLE / SLA Review
- Questions & Feedback

Production Management

- Daily Standups
- Issues & Alerts
- Promotion
- Training & Support

Learn

Continuous Improvement

- Ideas
- Suggestions
- Improve Value:
- - Process
- - People
- - Technology

Lessons Learned

- Issue Mgmt.
- Change Mgmt.
- Consumer Mgmt
- Production Mgmt.
- Cont. Improvement

Workshop

Getting Started with Modeling a Data Product
Developing the Certification Plan & Engagement Model

Modeling a Data Product:

- Let’s outline all the components that we will need to model by leveraging the Data Product Concept we introduced earlier:
  - Stakeholder Value
  - Use Cases & Features
  - Product Design – Data Sets(s) to deliver to stakeholders
  - Product Build - Process Plan (recipe) & BoM (ingredients)
  - Product Quality – Measures, Data, Test Harnesses
  - Certification – Plan to ensure all the above was done RIGHT
  - Release – Plan to transition into Production
  - Product Operations – Measures, Processes, Artifacts
  - Continuous Improvement – Change Mgmt., Issue Mgmt., Prioritization

Setup Checklists for EACH Component:

Example: Certification

Data collected	Quality metrics defined	DSA and SLA exist	Security Certified	Privacy Certified	Metadata defined	Taxonomy defined	Ontology defined	Fit for purpose	Fit for publish
Y	N	N	N	N				N	N
Y	Y	Y	Y	Y				N	Y
Y	Y	Y	Y	Y	Y	N	N	Y	Y
Y	Y	Y	Y	Y	Y	N	N	Y	Y
Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Workshop Goals:

- Verify steps in the process plan
- For each step in the process plan:
  - Verify setup for each step
  - Verify inputs & outputs for each step (quality measures)
  - Verify ownership for each artifact
  - Verify components
  - Certify final product

Generic Process Plan for Building a Data Product

These steps are typically performed for every data product. The goal for the workshop is to confirm the roles (see RACI) and identify the artifacts & components required for each step.

Each step is broken down using SIPOC model inputs, process, outputs so we can clarify the configuration and setup for processing a generic product in the Data Factory.

Process Plan Step: Acquire

Core Artifacts:
- Data Sources
- Data Types
- Data Quality
- Catalog

Attendees:
- ✔ Strategy & Planning
- ✔ Architecture & Products
- ✔ Governance & Policy
- ✔ Data Operations
- ✔ Data Engineering
- ✔ ML Engineering
- ✔ Platform Engineering
- ✔ Core Models
- ✔ Enablement
- ✔ Analytics & Adaptive Models

Goals:
- **Inputs**
- **Processing**
- **Outputs**
- **Measures**

Quality Artifacts:
- Policy
  - Rules / Constraints
- Security
  - Measures
  - Classification
  - Protection
  - Provisioning
  - Masking

Engagement Model:
- **Roles**
- **Activities**
- **Communication**
- **Orchestration**

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Process Plan Step: Clean

Core Artifacts:
- Data Sources
- Measures
- Data Profiling
- Data Cleansing / Alignment, Formatting

Attendees:
- ✔ Strategy & Planning
- ✔ Architecture & Products
- ✔ Governance & Policy
- ✔ Data Operations
- ✔ Data Engineering
- ✔ ML Engineering
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Process Plan Step: Transform

Core Artifacts:
- Catalog
  - Taxonomy
    - Classifications
- Transforms
  - Mapping
  - Masking
  - Merging
  - Improving
  - Integration
- Measures
  - Data Profiling
  - Data Quality

Attendees:
- ✔ Strategy & Planning
- ✔ Architecture & Products
- ✔ Governance & Policy
- ✔ Data Operations
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Goals:
- **Inputs**
- **Processing**
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- **Measures**

Engagement Model:
- **Roles**
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Process Plan Step: Certify

Core Artifacts:

- Data Sources
- SLAs (Vendors, Providers, Partners)
- Discovery (Metadata, Profiling)
- Models (Taxonomies, Ontologies)
- Transforms (Map, Merge, Repair, Replace)
- Policies (Rules / Constraints)
- Protection (Access, Masking, Privacy)
- Quality (tolerances, acceptance, auditing)
- Measures (Metrics, Trending, Projections)
- Catalog

Attendees:

✔ Strategy & Planning
✔ Architecture & Products
✔ Governance & Policy
✔ Data Operations
✔ Data Engineering
✔ ML Engineering
✔ Platform Engineering
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✔ Analytics & Adaptive Models

Goals:

- **Inputs**
- **Processing**
- **Outputs**
- **Measures**

Engagement Model:

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Process Plan Step: Publish

Core Artifacts:
- Fit for Purpose
- Fit for Quality
- Fit for Protection
- Fit for Policy
- Fit for Operations
- Fit for Resiliency
- Fit for Auditability
- Fit for Support
- Fit for Reusability
- Fit for Releasability
- Fit for Maintainability

Attendees:
- ✔ Strategy & Planning
- ✔ Architecture & Products
- ✔ Governance & Policy
- ✔ Data Operations
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Goals:
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Plan | Partner | Participate | Orchestrate

Future Planning & Opportunities

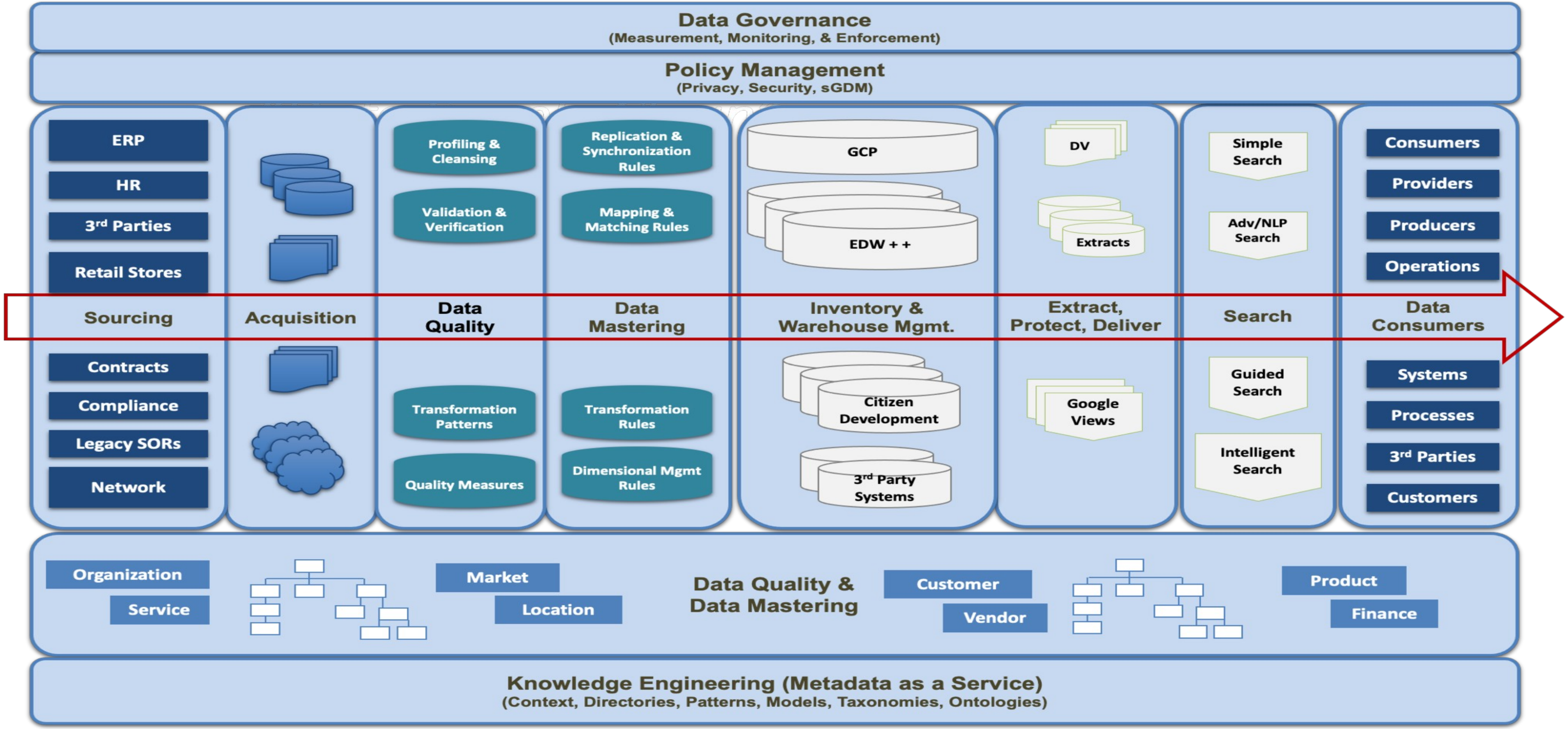
Future of Growth

- Sharing
  - Silos are based on islands of control and have limited value and opportunity in the future. The digital ecosystem(s) that are evolving rely upon shared values and participatory models of cooperation and collaboration.

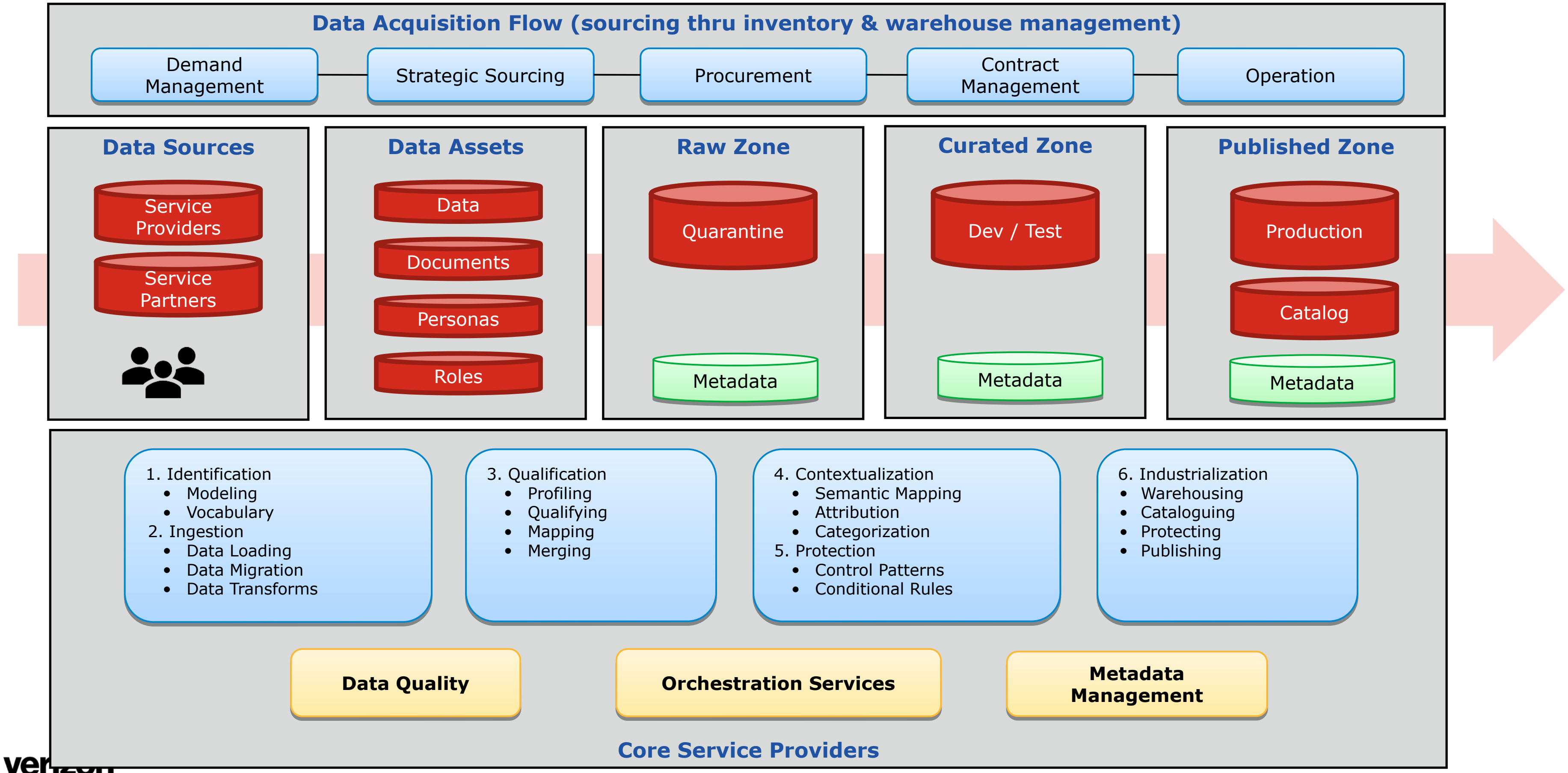
- Ecosystems
  - Digital twins for configuring and maintaining systems, equipment, and components.
  - Digital knowledge fabrics for planning, managing, and delivering services for operations.
  - Lifecycle management and Change Management requires orchestration.
  - Identity needs to be managed using a federated registry (which is block chained)
  - Security is more of an extension via communications (designed as a fabric not rules)
  - Digital Marketplace will provide visibility & availability for all products and services

Business Model

Digital Supply Chain (Data Flow)



Data Acquisition & Onboarding (Applying DCAM)



Data Supply Chain **Future State**

Providers
(Internal SORs)
(3rd Parties)

Material
(Data Acquisition)

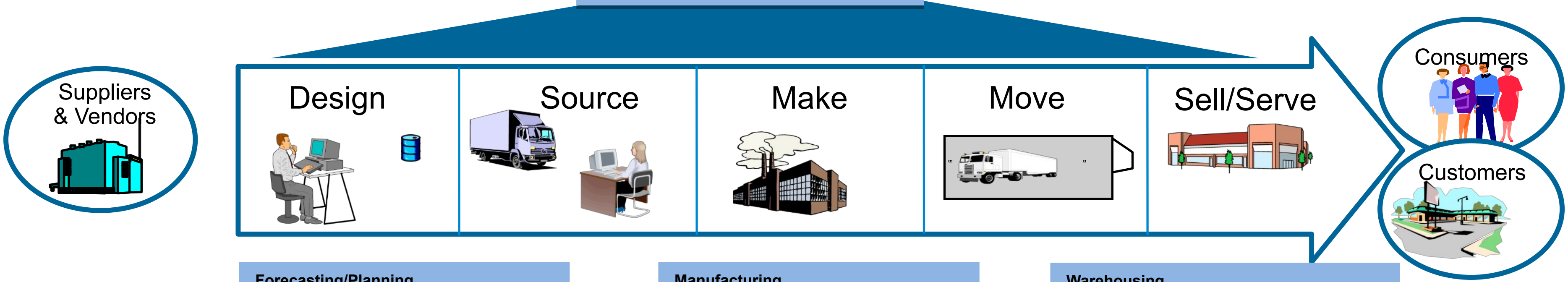
Product
(PLM / Certify)

Portal
- Search, Sample, Subscribe
- Order Management
- Request Management
- Operations Management
- Measure, Monitor, Manage

Market
(Publish via Portal)

Package / Label
(Catalog / Classify)

Consumer



Forecasting/Planning
- Demand Management
- Supply Planning

Sourcing
- Supplier Qualification
- Proposal Management
- Negotiation/ Contracting
- Supplier Administration

Procurement
- Planning
- Purchasing
- Purchasing Administration
- Settlement/Payment

Manufacturing
- Make-to-Stock
- Order-to-Stock
- Make-to-Order

Material Acquisition
- RAW Data
- New Products

Knowledge Mgmt.
- Contextualization
- Structured Vocabularies
- Model Management
- Ontology Management

Warehousing
- Storage Management
- On Prem / Cloud Services
- Data Modeling & DCAM
- Leaf / Streamed / Aggregate
- Structured & Unstructured
- Data Quality & Mastering

Extraction & Delivery
- Data Integration Layer
- Data Virtualization
- Data Integrity & Availability
- Data Privacy & Security
- Data Releasability
- Data Analytics & Algorithms

Service Management

Service Providers deliver their value in both Projects and Production Support.

The processes and deliverables are the same.

The measures (SLEs) are not and that is an important differentiation.

Lifecycle Management

- Engagement
  - Planning
  - Define Requirements
  - Design Review
- Estimation
- Service Delivery*
- SLEs for Lifecycle Delivery

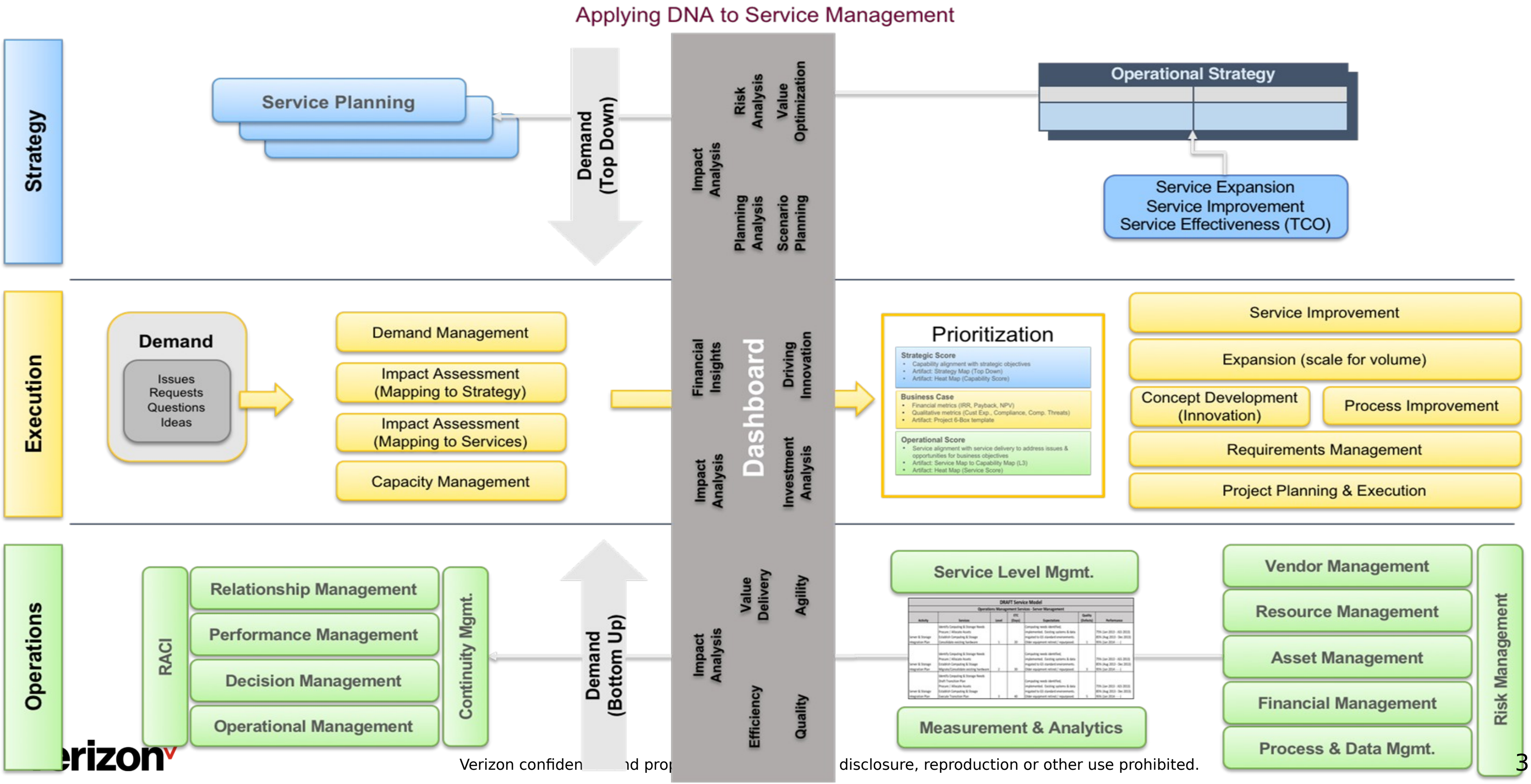
Operations Management

- Request Management
  - Ad Hoc
  - Consultation
  - Operations Support
  - Demand Planning
- Service Delivery*
- SLEs for Operations Delivery

*Service Breakdown

- Processes
- I-P-O
- Value Delivery

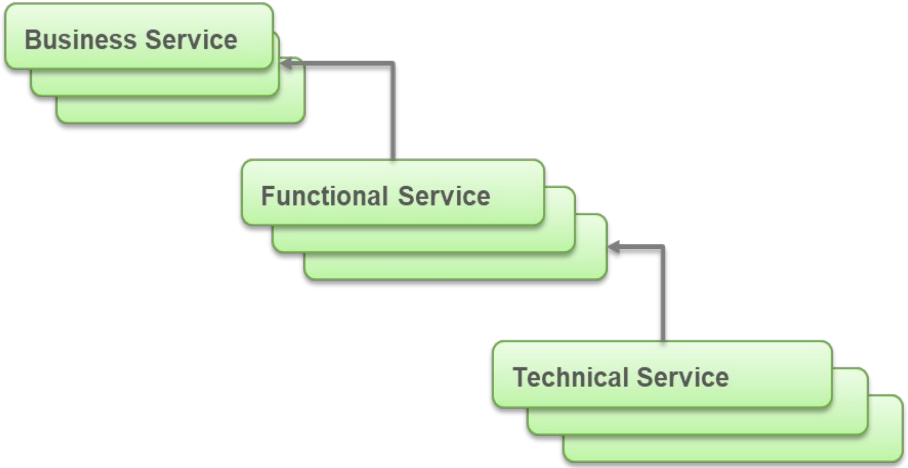
Operational Model



Optimizing Value Delivery with Services

This is where value is created and delivered. The capabilities embodied in these areas provide the foundation for the business architecture.

All other functional operations exist to guide and support these 4 capabilities. They provide services to enable these capabilities and support other functional areas.



Capabilities vs Services vs Processes

- **Capabilities**
  - Defines the (WHAT) for an organization does to create & deliver value
  - Align directly with Strategy (WHY) & Operations (HOW)
  - There is no ownership of Capabilities
  - Provides a way of contextualizing value delivery from the Organization’s perspective
  - Provides a clearly defined boundary for the value domain

- **Services**
  - Are consumed by one or more stakeholders (internal and/or external)
  - Directly support Capabilities
  - Are owned, measured, and managed by a ”named” stakeholder
  - Are Customer Focused
  - Performance is measured by Customer Feedback
  - Provide a clearly defined boundary for the problem domain *

- **Processes**
  - One or more activities that are used to deliver a service
  - An Activity has inputs & outputs and are initiated by a trigger (planned or unplanned event)
  - An Activity is (typically) performed by one stakeholder (individual) with specific skills / knowledge
  - A Flow is a series of Activities (and Events) that are performed in a defined sequence
  - A Process is a Flow with a clear START & END points
  - Processes are execution focused
  - Performance is measured by Accuracy & Consistency

Capabilities	Capabilities
• Customer Relationship Mgmt. • Strategic Marketing • Marketing Communications • Sales Operations • Commercialization	• Planning, Governance, & Control • Operations Management • Financial Management • Resource Management • Project Management • Issue Management

Inputs (Services)	Processes	Outputs (Services)
• Strategic Management • Product Management • Supply Chain Management • Contract Management • Competitive Intelligence • Communications Mgmt.	• Perform Customer Focus • Evaluate & Prioritize Market Opportunities • Define Market Segments • Capture & Warehouse Desired Outcomes • Conduct Sales Budgeting • Manage Sales Operations • Measure Gaps in Desired Outcomes	• Digital Marketing Services • Marketing Segmentation • Lead & Quote Management • Sales & Order Management • Customer Focus & Analysis • Innovation Management • Customer Loyalty Management • Service Level Management

* key for applying AI & ML effectively

Service Planning Model

Service Planning Model

A Service Can Be: Business, Functional, or Technical

- Capabilities are enabled by one or more services
- Each service is designed as a “mini-supply chain”
- Service Level Management is designed to ensure operational effectiveness
- Service Ownership is where we simplify Accountability (instead of relying on complex RACI models for resources)
- Measures quality, productivity, and service excellence (what gets measured gets done right)
- Service has its own business framework (revenue, cost control, measurement, supplier and customer management)
- Policies impose constraints on processes and they may require additional desired outcomes (which opens the door for innovation)

Example: Technical Service Model

Service Planning Model

Technical Service L1	Technical Service L2	List of Processes
Operations Management	Computing Service	Computing Environment Provisioning & Maintenance Server & Computing Cycle Planning & Maintenance Technical Architecture Deployment Distributed Computing Support (includes Utility Service Delivery)
Operations Management	DC Operations Service	Monitoring & Maintenance of Computing Facilities Installation & Upgrade of computing platforms Coordination with Continuity Planning teams Provide proactive monitoring & management with other service owners (availability & planning, procurement, & vendor mgmt.)
Operations Management	Event Management Service	Messaging layer services for SOA Workflow (OLTP, manual workflow, distributed, and slow-moving workflow) Auditing & Instrumentation of all events and activities to monitor for anomalies and conduct proactive trending to anticipate problems
Operations Management	Messaging & Collaboration Service	Email storage and provisioning Vendor Mgmt. and license mgmt. Distribution at the desktop, mobile, or web

Service Measurement

DRAFT Service Model						
Operations Management Services - Server Management						
Activity	Services	Level	ETC (Days)	Expectations	Quality (Defects)	Performance
Server & Storage Integration Plan	Identify Conputing & Storage Needs Procure / Allocate Assets Establish Computing & Stoage Consolidate existing hardware	1	20	Computing needs identified, implemented. Existing systems & data migated to GS standard environments. Older equipment retired / repurposed.	1	75% (Jan 2013 - JGS 2013) 85% (Aug 2013 - Dec 2013) 95% (Jan 2014 - - -)
Server & Storage Integration Plan	Identify Conputing & Storage Needs Procure / Allocate Assets Establish Computing & Stoage Migrate/Consolidate existing hardware	2	30	Computing needs identified, implemented. Existing systems & data migated to GS standard environments. Older equipment retired / repurposed.	3	75% (Jan 2013 - JGS 2013) 85% (Aug 2013 - Dec 2013) 95% (Jan 2014 - - -)
Server & Storage Integration Plan	Identify Conputing & Storage Needs Draft Transition Plan Procure / Allocate Assets Establish Computing & Stoage Execute Transition Plan	3	40	Computing needs identified, implemented. Existing systems & data migated to GS standard environments. Older equipment retired / repurposed.	5	75% (Jan 2013 - JGS 2013) 85% (Aug 2013 - Dec 2013) 95% (Jan 2014 - - -)

Services used to support Data Marketplace

- **Support Services**
  - Big Data & Analytics
  - Storage & Compute
  - DevOps
  - Interface
  - Visualization

- **Financing & Payment Services**
  - Deployment
  - Working Capital
  - Insurance
  - Risk & Trust

- **Orchestration Services**
  - Registration
  - Integration
  - Interoperability
  - Security

- **Infrastructure Services**
  - Equipment & Asset Mgmt.
  - Communications Management
  - Power Management
  - Network & Systems Mgmt
  - Fiber & WIFI Management

- **Supply Chain Services**
  - Procurement Management
  - Operations Management
  - Product Management
  - Distribution Management

Provider — Partner— Consumer Marketplace

- **Virtual Provider**
  - Pricing & Support
  - Billing, Payment
  - Service Levels
  - Systems & Information
  - Services (Business, Functional, Data)
- **Virtual Operator**
  - Monitoring & Escalation
  - Consumer Support
  - Change & Lifecycle Management
  - Measurement
  - Service Level Management
  - Incident Management
  - Issue Investigations & Resolution

Digital Services Portal offerings

Digital Marketplace Home
- Catalog
- Ecosystem Orchestrator
- Marketing & Sales

Wholesale Communications
- Carrier Ethernet
    - Small Cell Neutral Hosting
- Network Access and Backhaul
    - Fixed Wireless Access
    - Direct Internet Access
    - Cellular Neutral Access Hosting
- Voice Services
    - Unified Communications
- Content Services
    - Streaming Broadcast
    - On Demand
    - Learning

Power Management
- Building
- Microgrid as a Service
- Transactive Energy

Public Safety as a Service
- Emergency Operations Center
- Incident Management
- Resource Management
- Radio Control
- Video Surveillance

Building Automation
- Exterior and Interior Systems
- Building Access
- HVAC
- Elevators
- Digital Signage

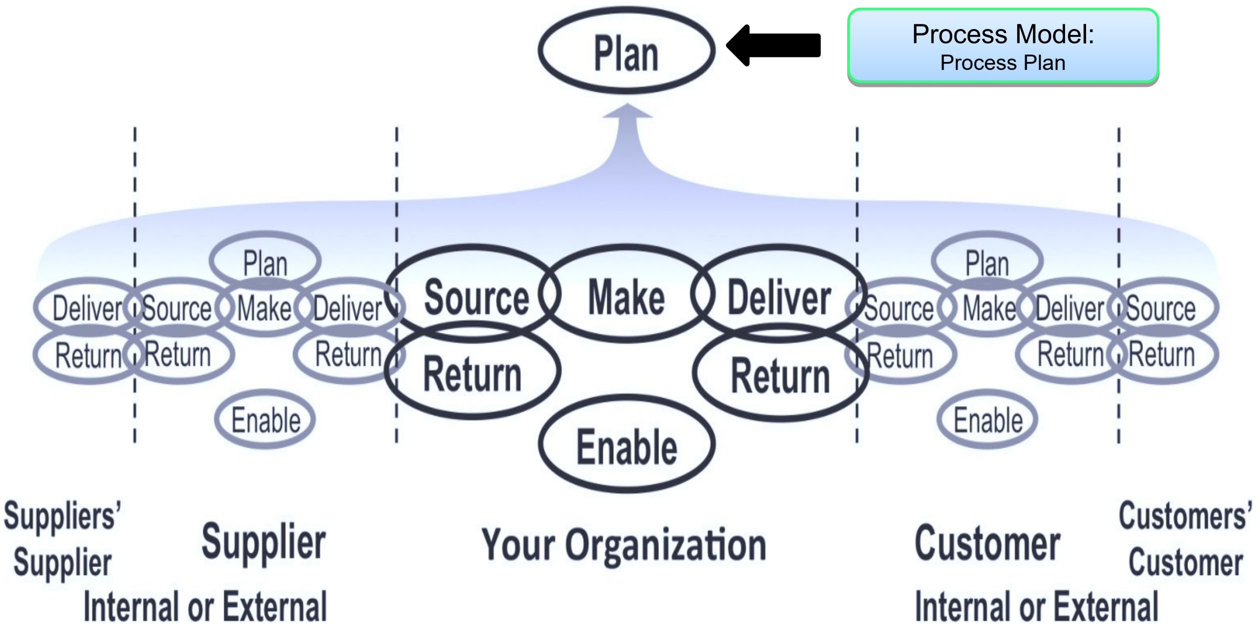
Concierge Services
- Systems Administration
- Cybersecurity

Management as a Service
- Situational Awareness
- Ops Center
- Fault Management
- Configuration as a Service
- Billing as a Service
- Performance Mgmt. as a Service
- Precision Time as a Service

Security as a Service
- SIEM
- ThreatConnect
- Notifications
- Cell Broadcast

Leveraging SCOR & PLM

Establishing the Supply Chain for Service Delivery

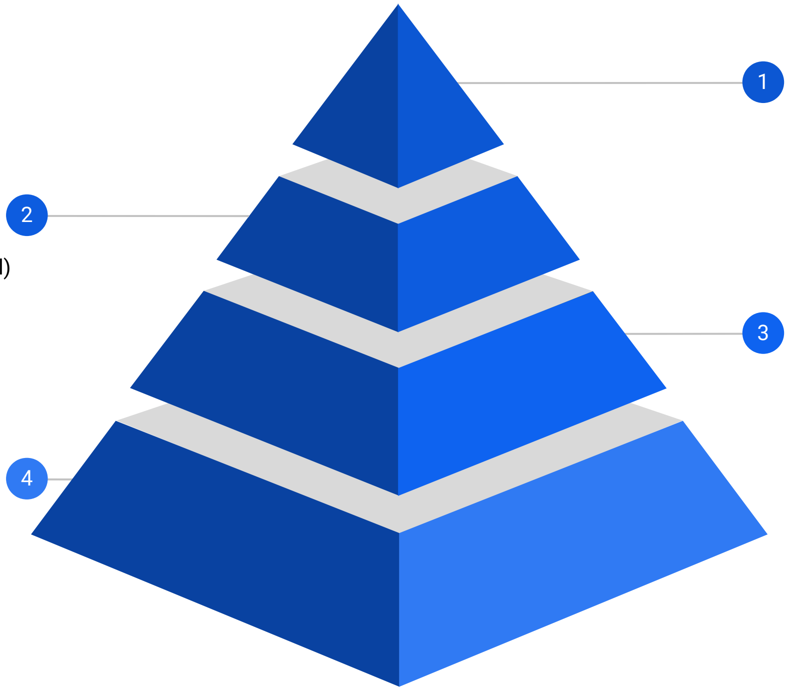


- SCOR Model provides a service vocabulary for the Data Factory
- PDMA provides a design and build vocabulary for a data product

Measurement

Performance Attribute	Definition
Reliability	The ability to perform tasks as expected. Reliability focuses on the predictability of the outcome of a process. Typical metrics for the reliability attribute include: On-time, the right quantity, the right quality.
Responsiveness	The speed at which tasks are performed. The speed at which a supply chain provides products to the customer. Examples include cycle-time metrics.
Agility	The ability to respond to external influences, the ability to respond to marketplace changes to gain or maintain competitive advantage. SCOR Agility metrics include Flexibility and Adaptability
Costs	The cost of operating the supply chain processes. This includes labor costs, material costs, management and transportation costs. A typical cost metric is Cost of Goods Sold.
Asset Management Efficiency (Assets)	The ability to efficiently utilize assets. Asset management strategies in a supply chain include inventory reduction and in-sourcing vs. outsourcing. Metrics include: Inventory days of supply and capacity utilization.

Analytic Pyramid



Solutions Built Atop Metrics

Embedded Metrics (USD or Executable) in processes
Callable Analytic Models
Analytic Products (immersive interaction)

User Defined Metrics

Upper Level Metrics built atop one or more Executable Metrics and/or Base Calcs
Customized Dimensions (Custom Data Model)
User Defined Rules (Additional Filtering)

Executable Metrics

Platform Specific Code (Ansi SQL)
Unified Model (Harmonized Structure)
Dimensions (Mapped to Unified Model)
Executable Rules (Constraints & Filtering)

Base Calcs

Mathematical expression (YAML)
Dimensions (Standard Hierarchies)
Rules (AND not OR)

Reference

How do we specify a "generic" Data Product?

- identifier
- name
- description
- properties
- owner
- publisher
- features
- semantics
- services
- fit for purpose

- consumption modality (pub/sub)
- consumer (types)
- constraints (temporal)
- constraints (policy)
- constraints (protections - security)
- constraints (limitations or boundaries of performance)

These are some of the features used to define a "generic" Data Product

Components Required to Build a Generic Data Product

Data Acquisition (collect)

- Data Sources
  - Internal / 3rd Party
- File Types
  - System Feed
- Feed Types
  - In Motion / Streamed

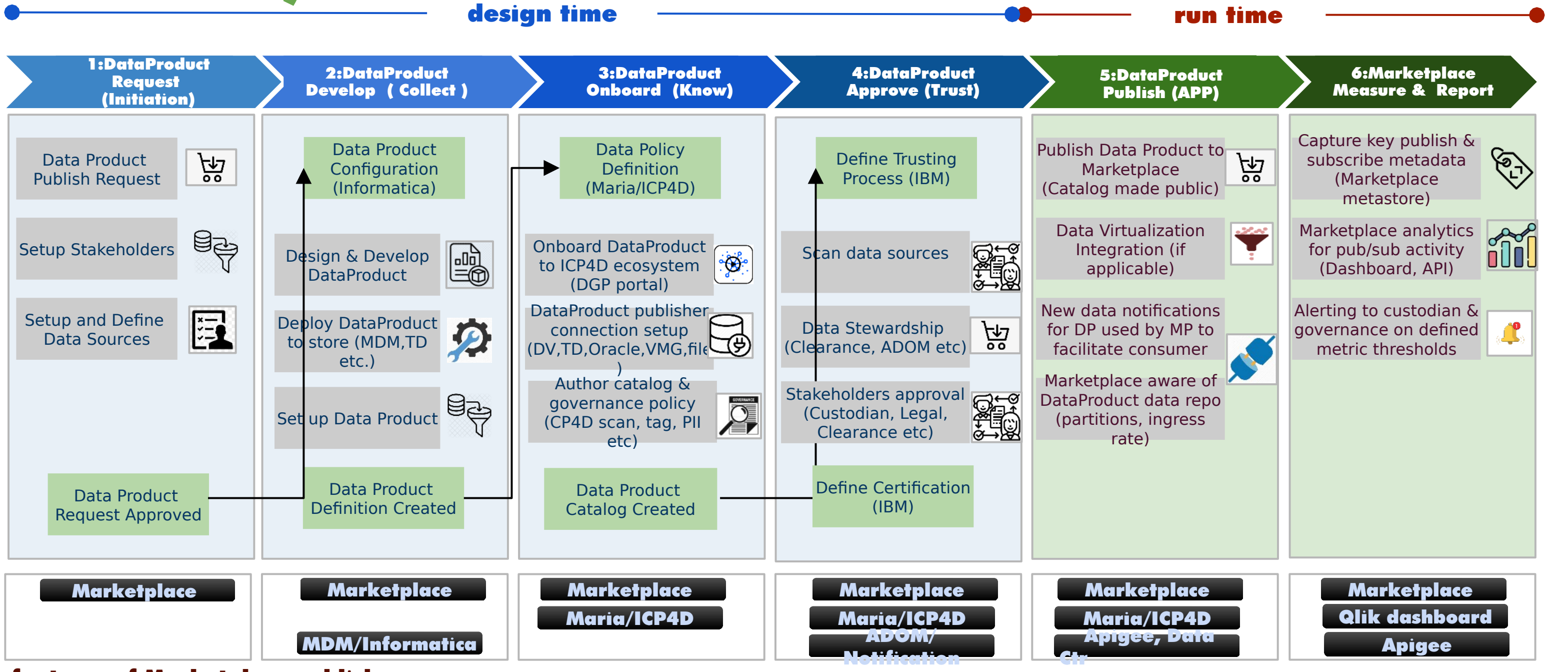
Data Insights (Know)

- Taxonomy
  - Domains, Entities
  - Classifications, Terms
- Ontology
  - Relationships/Lineage
  - Rules, Constraints
  - Policies / Controls
  - Standards/Guidelines
- Discovery
  - Automated
  - AI/ML Enabled
  - Metadata
- Catalog
  - Search
  - Metadata
  - Knowledge Base

Trust

- Transform
  - Mapping / Merging
  - Profile & Qualify
  - Contextual Alignment
  - Semantic Alignment
  - Metadata Enrichment
- Stewardship
  - Business Engagement
  - SLA/MoU/DSA
  - Remediation Workflow
- Data Quality
  - Measures / Tolerances
  - Traceability, Audit
  - Inbound data
  - Process steps
  - Outbound data
  - Context & Components
- Certification
  - Inputs / Process / Outputs
  - Process Artifacts
  - Policies / Quality Measures
  - Maturity Model
  - Risk Assessment

Data Marketplace Publisher Step Flow - (Greenfield)



Key features of Marketplace publisher model

- Self-serve secure platform to onboard backend data to consumer
- Avoid custom/manual processes to socialize data to consumer
- Define secure governance workflows centrally on datasets based on requirements
- Single source for all data governance policies
- Offer enterprise data catalog exploration on the web platform
- Uniform & Centralized control for data publisher on offering access to data on different platforms

#FORWARDTOGETHER